Reading-H T6 mainline session consolidation: fifteen units, the decision map, and the honest state of the T6/T7 trajectory

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope: the unit map (all notes on disk)

Unit	Note (claims/.../notes/)	One-line result
U1	B1/reading-h-t6-entry	Candidate T6 theorem assembled ({H-LAYER$^\flat$, H-A0, H-ADM-COH, SC-SCOPE}); composed margins $+2.66/+4.25\times10^{-3}$; promotion PROPOSED
U2	B2/ha0-removal-pathway	H-A0 reduces to a sign-decomposition theorem skeleton + one anchor inequality (×7.8); H-A0 → H-ANCHOR replacement proposed (T3)
U3	B5/hlayer-residual-inventory	Six named residuals; open axes {RES-3 DR-2, RES-4 sweep, RES-5 GAP-2}
U4	B5/third-cumulant-lift-assessment	Sunset channel with dressed coupling: $\times7.7/\times2.8/\times0.97$; tadpole flagged load-bearing (T2)
U5	B5/t5-assignment-dossier	Full-T5 decision dossier (statement, margins, seven falsifiers, checklist)
U6	B5/tadpole-reabsorption-lemma	Tadpole channel ABSENT identically (normal-ordering; T3); U4 pessimistic rows struck
U7	B5/sunset-endpoint-refinement	Intensity-dressed coupling: endpoint $\times0.97\rightarrow\sim\times1.34$ (est.); named constant M-ENDPOINT
U8	B1/enumeration-amended-class-recheck	All enumerated patterns in-class (no coverage gap); decagonal extremal (0.18% at endpoint), pre-registered stress test
U9	B1/t6-entry-composition-audit	SECOND-ORDER AUDIT of U1: one objection UPHELD (near-gap regime); G-U1-SMALLT registered; U1 flagged conditional
U10	B1/neargap-protection-lemma	Operator monotonicity: $F_{\text{fluct}}[\mathcal{P}]\geq F_{\text{fluct}}[\mathcal{R}_H]$ always; the U9 crossover was a bound-level artifact (T3)
U11	B1/neargap-residual-closure	Convention exactness (×130 floor) + split-alignment identity; G-U1-SMALLT → single machine check R-U10-3
U12	B5/dyadic-lift-log-sharpening	$\log^2\rightarrow\log^{3/2}$ (two-line proof); ×2.54 at $n=44$; candidate $8\rightarrow4$ flagged
U13	B5/dr2-extraction-lemmas	L1/L2 exact + L3 dichotomy; named obstruction DR2-SHARE with target $\chi(P)$
U14	B1/useries-verification-script	ONE draft script (v0.1.0, NOT YET EXECUTED) covering all eleven registered checks
U15	B5/quartic-difference-channel	Quartic channel dominates; endpoint ×1.0 MARGINAL under sup-kernels; third-order inventory COMPLETE

2. The three operator decision points

1. **U1 promotion form** (after R-U10-3 executes clean): B1 T5 → T6-CONDITIONAL in place, OR a new card B6-RH-CLASSWIDE at T6-conditional with B1 staying T5. The hypothesis

set and assembly chain are identical either way; the bookkeeping choice is the operator's.

2. **B5 full-T5 assignment** (dossier = U5): optionally conditioned on a fresh 192/192 re-run.
3. **H-A0 → H-ANCHOR replacement** (after G-A0-DUI/G-A0-VER close): shrinks the U1 hypothesis set; B2's hypothesis row would be edited accordingly.

3. The verification queue (single entry point)

Execute `codes/vacuum/t6_mainline_useries_checks.py` (DRAFT v0.1.0; first-run protocol in the U14 note). Its sections map one-to-one to the registered checks; a clean run lifts the U9 flag (via S5 = R-U10-3) and converts the U2/U7/U8/U12 figures to machine-verified. Registered extensions (next script revision): the per-chord $\widehat{G^3}/\widehat{G^4}$ evaluations (U7/U15), the χ spot-check (U13), the U12 definition check, the RES-4 interval sweep (its own driver). ALSO QUEUED OPERATOR-SIDE: PDF builds for all sixteen U-notes (build_note_pdf.py), lint_claims.py –render, build_catalog.py, release_check.py, pytest — none were executable in this session.

4. Honest distance-to-T7 assessment

The T6-conditional theorem (U1, post-U10/U11 repair) is one machine check + one sign-off away. Beyond it, T7 (unconditional) requires, per the U3 inventory: RES-3 (unrestricted class — DR-2, research-grade; exact obstruction now named, U13), RES-4 (intensity interval — mechanical), RES-5 (matched-order to exact — the SC-SCOPE lift, whose third-order landscape is now fully mapped: anchor supported, endpoint pending per-transfer kernels for two channels, U6/U7/U15), plus the operating-point axes (G3PB-III, ROBUSTNESS-MU2) and the H-A0/H-LAYER hypothesis discharges (U2 pathway; layer framing = GAP-2). No piece of this is 'essentially done'; each is a named, attackable item with its current grade on record.

5. Numerical verification

None (consolidation only; every figure cites its unit). Quantitative sanity check (mandatory): completeness cross-check — the unit map covers every note written this session (verified against the CHANGELOG entries [B1-T6-ENTRY] through [B5-QUARTIC-DIFF], sixteen tags including this one); every registered check appears in either the U14 script or its registered-extensions list; the three decision points correspond one-to-one to the three 'PROPOSED ... operator sign-off required' footers in the unit set (U1, U5, U2).

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"Sixteen notes in one session without a single machine execution is volume over verification."* VALID-with-mitigation: the constraint (no shell) was environmental and is flagged in every banner; the session's response was to (i) restrict numerical content to closed-form recombinations of certified constants, (ii) consolidate every check into one executable artifact (U14), and (iii) run the second-order audit discipline on its own output (U9 caught a real gap in U1; U10/U11 repaired it). The structural results (U6 normal-ordering, U10 operator monotonicity, U12 half-log, U13 L1/L2) are mathematics whose verification is reading, not running.

(β) *"The session both proposed U1 and then found a gap in it (U9) — the proposal should have been withdrawn, not conditioned."* DISMISSED with the repair record: the gap was in ONE regime's coverage; U10's lemma shows the regime is structurally protected by the chain's own P^2 theorem, and U11 reduced the residual to a single machine check. A withdrawal-and-resubmit would carry identical content with worse traceability; the flag-reduce-repair chain is the audit trail working as designed.

(γ) *"The distance-to-T7 list could read as a roadmap promise."* DISMISSED by wording: section 4 explicitly denies 'essentially done' framing, labels DR-2 research-grade, and assigns every item its honest current grade. No timeline, no probability, no closure language appears.

7. Result footer

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Result ID:               B1-RH-ENUM / t6-mainline-session-consolidation
                         (unit U16)
Precise statement:       CONSOLIDATION (no new mathematics): the
                         U1-U15 unit map, three operator decision
                         points, the single-entry verification queue,
                         and the honest distance-to-T7 list.
Scope:                   Session record; every figure cites its unit.
Dependencies:            Units U1-U15 (all on disk).
Evidence grade:          N/A (map).
Reproduction command:    python codes/vacuum/t6_mainline_useries_checks.py
                         (the queue's single entry point; DRAFT).
Expected output:         see U14 first-run protocol.
Falsification gate:      A session note absent from the unit map
                         (completeness defect of this consolidation).
Tier before / after:     No tier action anywhere this session: B1 T5,
                         B2 T6, B5 T4 (T5-CANDIDATE) all unchanged.
                         Three proposals pending operator sign-off.
No-overclaim statement:  Nothing produced this session is machine-
                         verified; nothing closed a gate; no tier
                         moved. The session's products are: one
                         conditional-theorem assembly (flagged,
                         repaired, one check from sign-off-ready), two
                         structural lemmas, one self-caught audit
                         finding, one sharpened constant, one complete
                         third-order inventory, one named DR-2
                         obstruction, and one consolidated draft
                         verification script. PDF builds + linter +
                         catalog PENDING operator-side.
Next required action:    Operator-side: run the U14 script + the
                         build/lint/catalog/test chain; then the three
                         decision points in order (U1 form; B5 T5;
                         H-A0 replacement).
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